Hints: Methods of proof – 2

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The naive statement $\sum_{k\leqslant n} 1/k^2 \leqslant 2$ does not close: from $S_n \leqslant 2$ you cannot conclude $S_n + \frac{1}{(n+1)^2} \leqslant 2$. With the bound $2 - \frac{1}{n}$, the step needs $\frac{1}{(n+1)^2} \leqslant \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$, which is true.

Hint 2. For the product $P_n = \prod_{k\leqslant n} \frac{2k-1}{2k}$, the weaker bound $P_n \leqslant \frac{1}{\sqrt{3n}}$ is also true, but its induction step fails. Check that $\frac{2n+1}{2n+2} \leqslant \sqrt{\frac{3n}{3n+3}}$ is false. With $\frac{1}{\sqrt{3n+1}}$, the step needs $\frac{2n+1}{2n+2} \leqslant \sqrt{\frac{3n+1}{3n+4}}$.

Hint 3. Square and clear denominators: $(2n+1)^2(3n+4) \leqslant (2n+2)^2(3n+1)$. Expanding gives $12n^3 + 28n^2 + 19n + 4 \leqslant 12n^3 + 28n^2 + 20n + 4$. The base case $n = 1$ holds with equality.

Problem 2

(orthogonal basis)

Hint 1. If $B \equiv 0$, any basis works. Otherwise there is a v with $B(v, v) \neq 0$, by polarisation (here char $\neq 2$ is used): $2B(x, y) = B(x + y, x + y) - B(x, x) - B(y, y)$.

Hint 2. Then $V = \langle v\rangle \oplus v^\perp$, where $v^\perp = \{w : B(v, w) = 0\}$ is the kernel of a non-zero functional, so it has dimension $n - 1$. By induction $B|_{v^\perp}$ has an orthogonal basis; add v to it.

(rank-one decomposition)

Hint 1. Let A have rank $r \geqslant 1$. Find a rank-one matrix R with $\operatorname{rank}(A - R) = r - 1$ and induct on the rank.

Hint 2. Choose a basis $c_1, \ldots, c_r$ of the column space. Each column of A is $\sum_i \lambda_{ij} c_i$, so $A = \sum_{i=1}^r c_i \rho_i^T$ with $\rho_i = (\lambda_{i1}, \ldots, \lambda_{in})$. As an induction: subtracting $c_r \rho_r^T$ leaves a matrix whose column space is spanned by $c_1, \ldots, c_{r-1}$. Alternatively, if $a_{ij} \neq 0$, the matrix $A - \frac{1}{a_{ij}}(Ae_j)(e_i^T A)$ has rank $r - 1$.

Problem 3

Hint 1. Statement: if finitely many convex sets $C_1, \ldots, C_m$ in $\mathbb{R}^n$ ($m \geqslant n + 1$) are such that every $n + 1$ of them intersect, then all of them intersect. Induct on m; the base case $m = n + 1$ is the hypothesis. The tool that uses the dimension is Radon's lemma: any $n + 2$ points in $\mathbb{R}^n$ split into two parts whose convex hulls meet.

Hint 2. For $m \geqslant n + 2$: by the induction hypothesis, for each i the family without C_i has a common point x_i. Apply Radon to $x_1, \ldots, x_m$ (in fact to $n + 2$ of them): $\{1, \ldots, m\} = I \sqcup J$ and a point $p \in \operatorname{conv}\{x_i : i \in I\} \cap \operatorname{conv}\{x_j : j \in J\}$.

Hint 3. For each k, either $k \notin I$, and then all x_i with $i \in I$ lie in C_k, so $p \in C_k$ by convexity; or $k \notin J$, and the same argument applies with J. Hence $p \in \bigcap C_k$. The dimension enters only through Radon's lemma ($n + 2$ points in $\mathbb{R}^n$ are affinely dependent). The number of sets enters through the induction.

Problem 4

Hint 1. $\sqrt{2}$: if $\sqrt{2} = p/q$ with q minimal, then $\sqrt{2} = \frac{2q-p}{p-q}$ with $0 < p - q < q$, which contradicts minimality. (Alternatively, $p^2 = 2q^2$ forces p and then q to be even.)

($x^4 + y^4 = z^2$)

Hint 1. Take a solution with z minimal. Then $\gcd(x, y) = 1$, and (x^2, y^2, z) is a primitive Pythagorean triple. Say x is odd: $x^2 = a^2 - b^2$, $y^2 = 2ab$, $z = a^2 + b^2$ with $\gcd(a, b) = 1$ and a, b of opposite parity.

Hint 2. From $x^2 + b^2 = a^2$ with x odd, b is even and a is odd: $x = c^2 - d^2$, $b = 2cd$, $a = c^2 + d^2$ with $\gcd(c, d) = 1$. Then $y^2 = 4cd(c^2 + d^2)$, and the factors c, d, $c^2 + d^2$ are pairwise coprime.

Hint 3. So each factor is a square: $c = u^2$, $d = v^2$, $c^2 + d^2 = w^2$. This gives $u^4 + v^4 = w^2$ with $w \leqslant a < z$, a smaller solution.

(fundamental theorem of arithmetic)

Hint 1. Existence: a minimal integer > 1 without a factorisation can be neither prime nor a product of two smaller numbers. For uniqueness, let n be the least number with two different factorisations $p_1 \cdots p_r = q_1 \cdots q_s$. Then no p_i equals any q_j, since otherwise you could cancel and get a smaller counterexample.

Hint 2. Say $p_1 < q_1$. Consider $n' = n - p_1 q_2 \cdots q_s = (q_1 - p_1) q_2 \cdots q_s$. Then $0 < n' < n$ and $p_1 \mid n'$. Factor $q_1 - p_1$ and compare with a factorisation of n' that contains p_1. By minimality these agree, so $p_1 \mid q_1 - p_1$, i.e. $p_1 \mid q_1$, which is a contradiction.

Problem 5

(basis)

Hint 1. Apply Zorn's lemma to the linearly independent subsets of V, ordered by inclusion. The union of a chain is again independent, because any finite subset of it lies in one member of the chain.

Hint 2. A maximal independent set S spans V: if $v \notin \operatorname{span} S$, then $S \cup \{v\}$ is independent, which contradicts maximality.

(two points on every line)

Hint 1. Well-order the set of lines as $(\ell_\alpha)_{\alpha < \mathfrak{c}}$, so that every initial segment has fewer than $\mathfrak{c}$ elements (the initial ordinal of the continuum). Build increasing sets A_α, no three points collinear, each of size $< \mathfrak{c}$, such that ℓ_β meets A_α in exactly two points for $\beta < \alpha$.

Hint 2. At step α: if ℓ_α already contains two points, do nothing. It cannot contain three. Otherwise it contains at most one point. Avoid the lines through two points of the current set: there are fewer than $\mathfrak{c}$ of them, none equals ℓ_α, and each meets ℓ_α in at most one point. Since ℓ_α has $\mathfrak{c}$ points, add one or two new points on it.

Hint 3. New points avoid all lines through two old points, so no three points become collinear, and ℓ_α now has exactly two points. Later steps never add a third point to ℓ_α. Take $A = \bigcup_\alpha A_\alpha$. A countable construction cannot work here, because there are $\mathfrak{c}$ lines.